# HEALTHCONNECT CLINIC: WEEK 8 TRANSITION & READINESS DOCUMENT

**Track:** Data Analytics

**Stage:** Week 8 (Final Integration, Presentation & Project Showcase)

**Collaborating Pod:** HC-POD 4

## SECTION 1: WEEK 7 → WEEK 8 TRANSITION

### 1. Main Week 7 Output

- The Week 7 Analytics Testing and Refinement Notebook.
- Verified testing records with automated unit test assertions.
- Refined visual charts with clinic benchmark lines.

### 2. Most Important Testing Result

- The baseline no-show rate was verified at exactly 51.15% across 4,737 active records.
- The test set from Data Science matched my population numbers within 0.2% to 1.7%.
- Model recall reached 99.8% on long lead times.

### 3. Major Issue Discovered During Testing

- Visual charts lacked clinic average benchmark lines.
- Stakeholders could not quickly spot high-risk groups.
- The predictive model produced 570 false positives at the 0.40 threshold.

### 4. What Was Refined or Improved

- Added a blue dashed clinic average line at 51.15% to the lead time chart.
- Added a red dashed service average line at 54.17% to the channel chart.
- Added direct data labels and fixed code warnings.
- Set a clinical policy: false positives only trigger helpful reminders.

### 5. Component Now Considered Ready

- The HealthConnect Final Analytics & Decision Support Package.
- All KPIs, risk segments and visual charts are tested and locked.

### 6. What Still Needs to Be Integrated

- Final speaking handoffs with Data Science.

### 7. Track(s) Involved in Final Integration

Data Science Track.

### 8. What I Intend to Demonstrate During the Final Presentation

- How long lead times drive no-shows from 29.5% up to 71.4%.

- Why SMS is the best reminder channel for high-risk visits.
- How analytics guided Data Science feature engineering.
- The 4 practical scheduling rules for clinic staff.

## SECTION 2: PART 1 - FINAL INTEGRATION READINESS

| Requirement | Your Response |
|---|---|
| **What is my final component?** | The HealthConnect Decision Support Package (KPIs, validated insights, benchmarked charts and business rules). |
| **What problem does it address?** | High missed appointment rates (51.15%) and wasted clinic appointment slots. |
| **What is its current status?** | 100% complete, fully tested and ready for integration. |
| **What did I improve during Week 7?** | Added baseline benchmark lines to charts, verified calculations with automated assertions and cleaned code warnings. |
| **Which other track(s) does it depend on?** | Data Science (for validation) |
| **Which track(s) depend on my output?** | Data Science (for validated risk thresholds) and Project Management (for executive findings). |
| **What output will I provide?** | Verified clinic KPIs, high-resolution charts, 4 scheduling rules and presentation slides. |
| **What output do I need from another track?** | Final master presentation flow and timing from Project Management. |
| **What evidence demonstrates readiness?** | 8 passed test cases in Week 7, automated unit assertions and cross-track Slack sign-offs. |
| **What limitations remain?** | The dataset is synthetic, excludes children under 18 and lacks real-time message delivery receipts. |